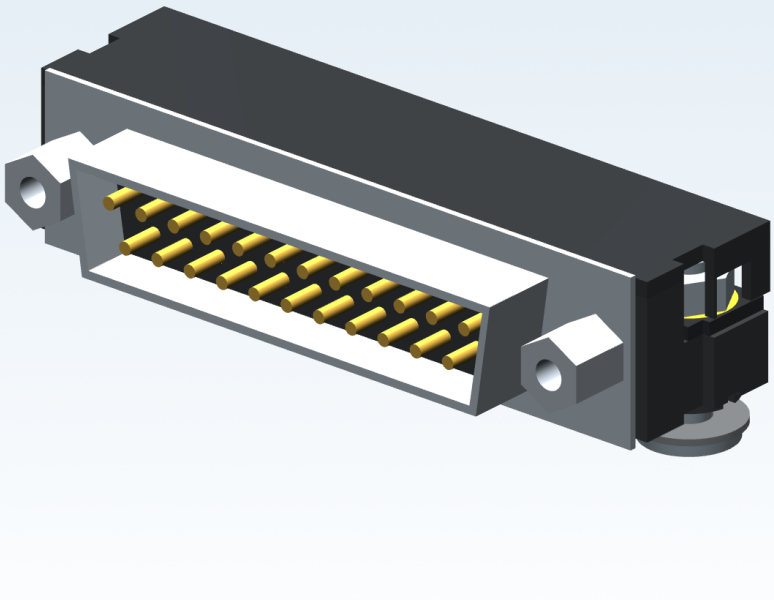
DB23 replacement assembly

Shorter mounting-ear PCB, solid printed base and black keyed clip-on cap

DB23 replacement - black clip-on cap fitted

UK Computing Limited | Revision 0.5 | CAD fit prototype



A solid base and two upward M3 screws provide the intended mounting; the cap retains jackpost nuts.
Unmeasured DB23 and bought-hardware reference shapes. Shell bonding wires are omitted.

The revision 0.5 assembly uses two metal screws from beneath the motherboard, through a solid base and the adapter PCB, with insulating washers below the motherboard and nuts above the adapter PCB. The cover conceals the mounting hardware and retains the jackpost nuts; it is not the primary cable-load attachment.

Adapter PCB overall width / depth	**51.20 / 9.00 mm (Y4.00-13.00)**
PCB thickness / underside height	**1.00 mm / Z5.75 mm assumed**
Printed base width / rear position	**52.00 mm / Y13.00 mm**
Printed cap width / rear position	**52.00 mm / Y13.20 mm**
Original shell mounting centres	**44.27 mm; Y9.50 mm**

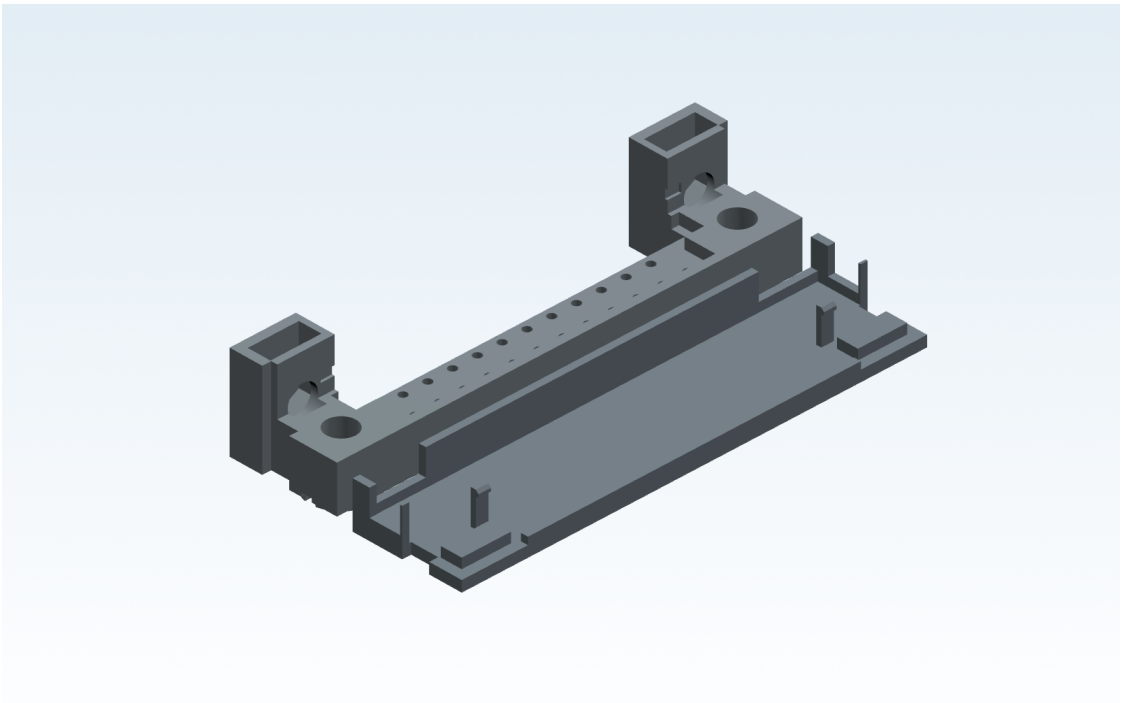
OEM comparison: iComp's original-equivalent drawing has a 50.27 mm width and 12.50 mm depth behind the flange datum. The present printed envelope is 52.00 mm wide and reaches Y13.20: +0.865 mm per side and +0.70 mm at the rear. The signal-hole pattern and nominal mating centreline are retained.

Fit prototype: purchased connector, printed fit, clamping, underside clearance and cable-load strength remain unverified. The DB23 body, cups and bought hardware in the views are explicitly assumed reference shapes.

The two printed parts

A solid base carries the PCB; a keyed cover clips onto the front towers

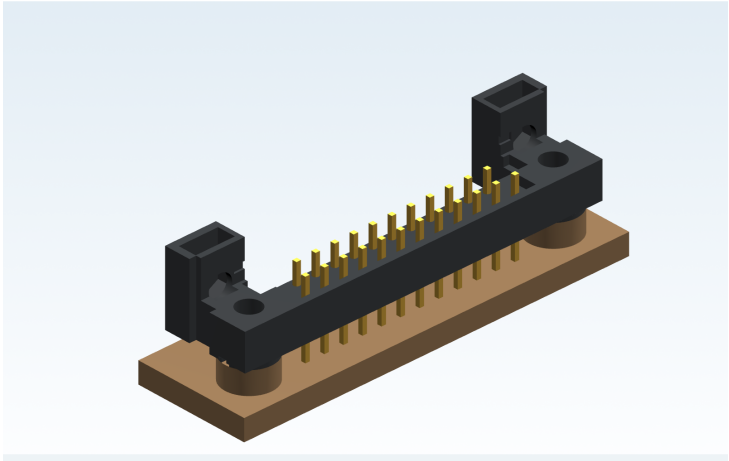
Printed base and keyed clip-on cap
UK Computing Limited | Revision 0.5 | CAD fit prototype



Actual printable geometry: base at its assembly position, cap in its roof-down print orientation.
Keyed contact faces locate the cover; flexible latches and nut-retaining tongues need a print-fit trial.

The base has continuous end supports and integrated feet at the original mounting positions. No separate 0.60 mm shims are used in this revision. Its 23-pin guide remains below the adapter PCB, leaving the nominal 1.25 mm solder-fillet and heat gap. J4 reliefs clear covered underside joints.

Pin loading - removable common-height jig
UK Computing Limited | Revision 0.5 | CAD fit prototype



Load equal-length pins against the lower-tip stop, then remove the jig before soldering.
The jig is a loading aid and is not fitted to the motherboard. Test pin retention after transfer.

Load the pins first

Print the hole-fit coupon and clip-fit pair before the full parts. The trial guide diameter is 0.92 mm for assumed 0.64 mm square pins; their diagonal is 0.905 mm. A finished 0.80 mm bore does not clear them.

Use the separate loading jig to set the lower tips. Equal 10.70 mm pins span Z-3.20 to Z7.50. Remove the jig before soldering and check that the base retains the pins.

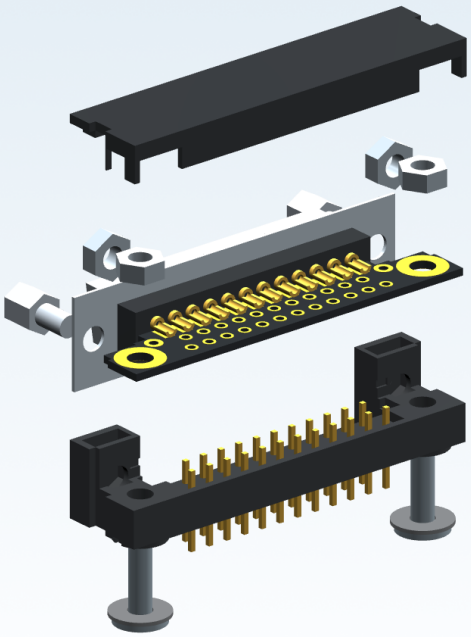
Printing: the base, cap, jig and hole coupon are trial parts. Follow the mechanical notes for print orientations and cap assembly transforms. The cap's flexible latches, nut tongues and keyed surfaces need an actual print-fit trial; material, soldering heat and long-term strength have not been validated.

Assembly in four parts

The DB23 and PCB stay joined; upward mounting screws are fitted during installation

Assembly - joined DB23 and PCB, upward screws, clip-on cap

UK Computing Limited | Revision 0.5 | CAD fit prototype



The DB23 and PCB move together. Two screws enter from below the motherboard; fit the cover last.
Offsets show part relationships, not a required insertion direction for every component.

Part 1 - Join the DB23 and PCB

Identify the matching male or female pin numbers. Solder both rows of cups to the PCB away from the printed base. Prepare the shell bond at the dedicated J4 pads and finish any covered underside joints.

Part 2 - Load pins, then lower the joined assembly

Load equal-length pins into the base using the height jig. Remove the jig, retain the pins in a heat-safe fixture and lower the complete DB23 + PCB from above. Seat the PCB fully, then solder all 23 pin tops.

Part 3 - Fit the jackpost hardware

Insert the jackpost nuts and fit the hex posts through the connector flange. Check actual flange bearing, nut fit and rear stem projection. Leave the decorative cap off while fitting and checking the mounting hardware.

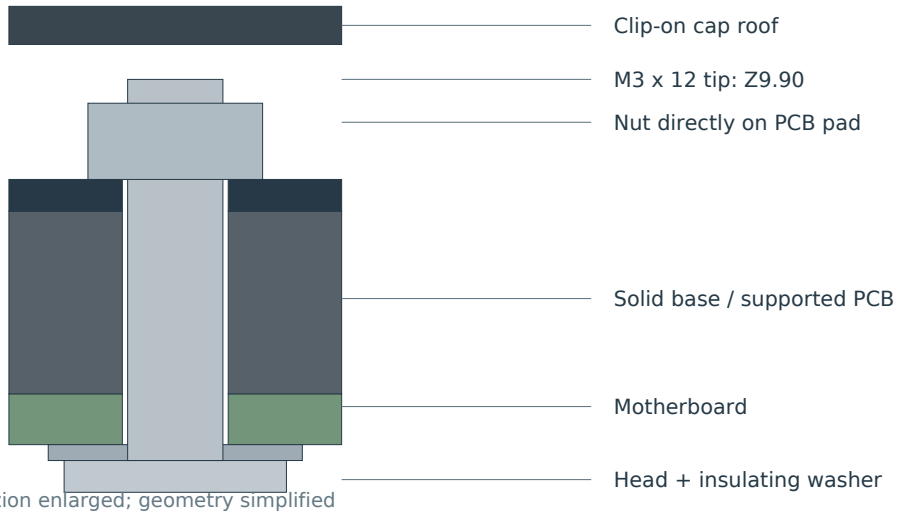
Part 4 - Mount, solder, check, then cap

Check all 23 circuits off the motherboard. Pre-tack the motherboard shell-pad lead with the base removed, then route it through the foot groove. Install screws upwards from below, with insulating washers below the motherboard and nuts on the PCB mounting pads. Secure the supported stack, solder the motherboard tails and complete the shell bond. Inspect and test before clipping the cover on last.

Upward screw mounting stack

Nominal dimensions only; the actual motherboard, washer and nut must be measured

Motherboard thickness assumed	**1.60 mm**
Motherboard top to PCB underside	**5.75 mm**
Adapter PCB thickness	**1.00 mm**
Underside insulating washer	**0.50 mm**
Nut height; no top washer	**2.40 mm**
Total under head to top of nut	**11.25 mm**



Local section enlarged; geometry simplified

Length selection

Screw length below head	**Projection past nut / tip height**
12.00 mm - default	**0.75 mm / Z9.90**
14.00 mm	**2.75 mm / Z11.90**
16.00 mm	**4.75 mm / Z13.90**

Cap clearance: the default 12 mm screw ends at Z9.90, below the nominal roof underside at Z11.00. Untrimmed 14 mm and 16 mm screws project into the cover envelope and are not default-compatible. Changing the insulating washer, nut, board thickness or cap geometry requires rechecking the complete stack.

Original holes only: the 7 mm head sits over an 8 mm insulating washer. Allow 1.50 mm total projection below the motherboard. Verify chassis and copper clearance; the shell bond remains a separate wire connection. The washer pair spans 52.27 mm: 0.135 mm beyond each side of the 52 mm print. Do not countersink the motherboard. Wire routing and actual chassis clearance still need a fit check.

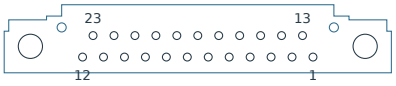
Pin templates and fit worksheet

Print this page at 100% / Actual size with no page scaling

Motherboard top view; connector faces the top of the page

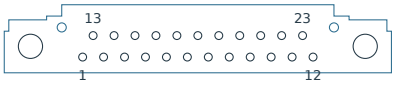
Signal holes and shell anchors retain the original nominal grid. J4 belongs to the adapter only.

MALE / RGB



Blue: adapter outline and J4 holes

FEMALE / FLOPPY



Blue: adapter outline and J4 holes

50.00 mm scale check

Measure before fabrication or final assembly

Purchased DB23 part / gender	_____
Cup gap / cup length / body depth	_____
Actual motherboard / PCB thickness	_____
Pin section / finished bore / retention	_____
Jackpost rear thread / stem projection	_____
Nut AF / thickness / pocket fit	_____
Upward screw / washer / nut stack	_____
Head bearing / underside chassis gap	_____
Cap latch fit / removal / tip clearance	_____
Shell bond / clamping / insertion trial	_____

Coordinate system: X runs across the connector; Y points inward from the flange front plane; Z is measured upward from the motherboard top. Male pin 1 is at positive X in this top view; female numbering is mirrored. The pin pitch is 2.77 mm, row spacing 2.84 mm and row stagger 1.385 mm.

Sources: original nominal connector grid and envelope: iComp DB23 replacement drawings. All selected design choices and unmeasured envelopes are in source/design_parameters.json. See docs/VALIDATION.md for actual saved-file checks and the physical work still outstanding.